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The derivative $f'(x)$ is the instantaneous rate of change.

Geometrically it is the slope of the tangent line.

Power rule: $d/dx x^n = n x^{(n-1)}$.

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Sum rule: $(f + g)' = f' + g'$.

Product rule: $(fg)' = f'g + fg'$.

Chain rule: $d/dx f(g(x)) = f'(g(x)) g'(x)$.